MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY:10545670

(BALLISTIC CORRECTION M220/M246E3)

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Ballistic Correction Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

10545425	Sight Current Generator
10545670	Circuit Card Assembly, Ballistic Correction M220/M246E3

PACKAGING DATA SHEET

10545670	Packaging of Circuit Card Assembly, Ballistic Correction

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Item definition. This assembly is a double ended printed circuit board consisting of two capacitors and two sets of 14 resistors. The assembly is used to provide two different scaling factors within the Sight Current Generator Drawing Number 10545425 for different ammunition characteristics.

3.1.1 Major component list. The components of this assembly shall be as specified on the Engineering Parts List (EPL) 10545670.

3.1.2 Government furnished property list. Not applicable.

3.2 Characteristics

3.2.1 Performance. The assembly shall have the resistance and capacitance values specified in Table I for the M246E3 plug, and Table II for the M220 plug.

Table I. Capacitance and Resistance Values for M246E3 Connector

Between		Minimum Reading		Maximum Reading	
Pin	Pin	Ohm	Microfarad	Ohm	Microfarad
22	23	0.000		0.500	
19	20	5,890,000		6,510,000	
19	28	- -	0.009	- -	0.011
18	17	167,530		167,870	
16	15	929,070		930,930	
14	13	407.99		408.81	
12	13	583.51		584.59	
12	11	946.25		948.15	
10	11	1,814.1		1,817.9	
10	9	5,049.9		5,060.1	
8	9	52,297		52,403	
7	6	8,816.1		8,833.9	
5	6	2,544.4		2,549.6	
5	3	1,152.8		1,155.2	
2	3	690.61		691.99	
2	4	781.12		782.68	

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Table II. Capacitance and Resistance Values for M220 Connector

Between		Minimum Reading		Maximum Reading	
Pin	Pin	Ohm	Microfarad	Ohm	Microfarad
24	23	0.000		0.000	
20	19	5,890,000		6,510,000	
28	19	- -	0.009	- -	0.011
18	17	167,530		167,870	
16	15	929,070		930,930	
14	13	1,572.4		1,575.6	
12	13	1,403.6		1,406.4	
12	11	3,175.8		3,182.2	
10	11	14,256		14,284	
8	9	21,199		21,241	
8	7	3,763.2		3,770.8	
6	7	1,548.4		1,551.6	
6	5	855.94		857.65	
3	5	513.98		515.02	
3	2	361.84		362.56	
4	2	109.39		109.61	

3.2.1.1 Environmental. The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristic. The physical characteristics shall be as specified in drawing 10545670 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings. This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 10545670 and associated lists.

3.3.2 Standards of Manufacture. The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship. The workmanship shall be as specified in specification MIL-C-50785.

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4. QUALITY ASSURANCE PROVISION

4.1 General.

4.1.1 Responsibility for inspection. The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination. The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment.

4.2.1.1 Digital ohmmeter. The digital ohmmeter required should have a resistance readout to five digits from 100 ohms to 10,000,000 ohms with 0.01 percent accuracy.

4.2.1.2 Capacitance bridge. The capacitance bridge required should have a capacitance readout from 0.001 microfarad to 10 microfarads with one percent or better, accuracy and resolution.

4.2.2 Functional tests. Using the test equipment of 4.2.1, verify the resistance and capacitance values listed in Tables I and II of 3.2.1 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC form 1029, 10545670.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

Custodian
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Preparing Activity
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Project No. 1285-A311

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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